

## Retail Performance & Sales Analytics Dashboard(2011).



This dashboard provides a comprehensive, high-level overview of global transactional data, tracking key performance indicators (KPIs) across revenue, order volumes, and customer acquisition.

22K

Total Orders

8.30M

Total Revenue

5M

Total Units Sold

4K

Unique Customers

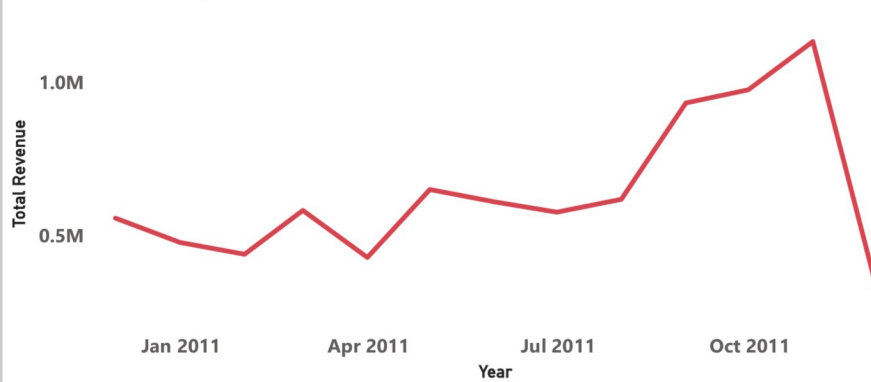
374.05

Avg Order Val

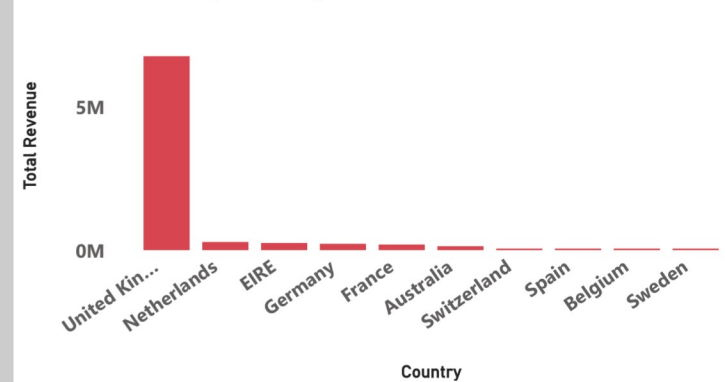
### Country

- ☐ Australia
- ☐ Austria
- ☐ Bahrain
- ☐ Belgium
- ☐ Brazil
- ☐ Canada
- ☐ Channel Islands
- ☐ Cyprus

Total Revenue by Year and Month



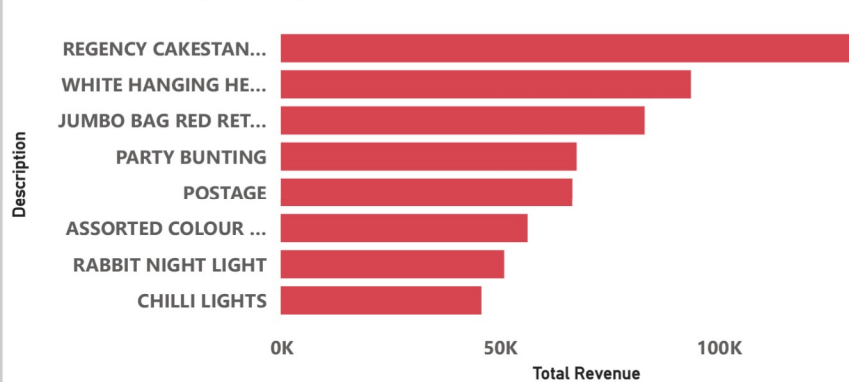
Total Revenue by Country



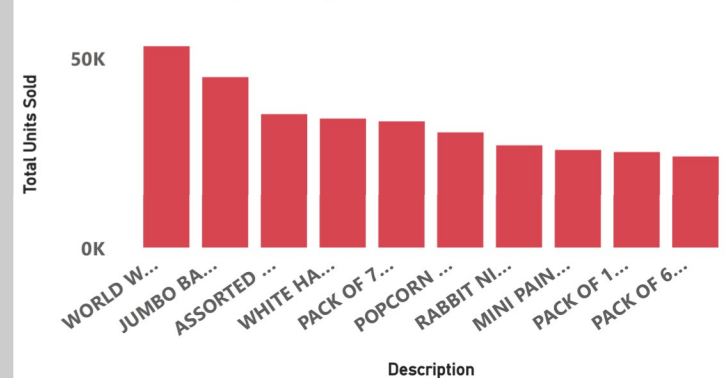
### Description

- ☐ ALARM CLOCK ...
- ☐ ASSORTED COL...
- ☐ BLACK RECORD ...
- ☐ CHILLI LIGHTS
- ☐ DOORMAT KEEP...
- ☐ DOORMAT RED ...
- ☐ HEART OF WICK...
- ☐ HEART OF WICK...

Total Revenue by Description



Total Units Sold by Description



### Customer Segmentation Methodology

To identify distinct buying behaviors within our transactional data, we applied an unsupervised K-Means Clustering algorithm based on engineered RFM (Recency, Frequency, Monetary) metrics.

- The Elbow Method (Left): We calculated the Within-Cluster Sum of Squares (WCSS) across various values of  $k$ . A clear inflection point or "elbow" is observed at  $k = 4$ , indicating the optimal number of segments before experiencing diminishing returns in variance explained.
- PCA Cluster Visualization (Right): Because the RFM feature space is multidimensional, we used Principal Component Analysis (PCA) to reduce the data into 2D space (PC1 and PC2). The resulting scatter plot validates that the four customer segments are distinct, well-separated, and mathematically cohesive.